LAN Simulation Project Documentation
Version 2.0: Enhanced Network with Services

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Version Note: This document describes **Version 2.0** of the LAN Simulation Project, which extends the basic VLAN implementation from Version 1.0 with additional enterprise network services: DHCP, wireless networking, email services, and enhanced DNS configuration.

Project Overview

This project involves the design and simulation of a multi-department Local Area Network using Cisco Packet Tracer. The network implements Virtual Local Area Networks (VLANs) to separate four academic departments while enabling controlled communication through inter-VLAN routing. This project is based on concepts from the research paper: "Design and Simulation of Local Area Network Using Cisco Packet Tracer."

Project Versions

Version	Core Features	Enhancements Over Previous Version
v1.0	Basic VLAN segmentation, Inter-VLAN routing, Static IP addressing	Initial implementation
v2.0	All v1.0 features plus: DHCP server, Wireless access, Mail server, Enhanced DNS	Added network services and automation

Project Repository

The complete project files, configurations, and additional documentation are available on GitHub:

GitHub Repository: `https://github.com/ZiadzWael/Design-and-Simulation-of-Local-A`

Version Structure:

- `/v1/` - Basic VLAN implementation (static IPs, no services)
- `/v2/` - Enhanced implementation with all network services

Project Team

Name	Student ID	Primary Responsibilities
Ziad Wael	221001480	Network architecture, VLAN configuration, router setup
Sama Eldessouky	221000579	Documentation, testing, IP addressing scheme
Abdullah Ramzy	221001781	Physical topology, switch configuration, validation

Academic Supervision

Dr. Islam Tharwat Abdelhalim
Project Supervisor and Course Instructor

Project Description

This project designs and simulates a functional Local Area Network for a university faculty with multiple departments. **Version 2.0** extends the basic VLAN implementation with enterprise-grade network services. The core concept implements Virtual Local Area Networks (VLANs) to create logical separation between different academic departments while maintaining controlled inter-department communication through inter-VLAN routing. The network follows a hierarchical design with access, distribution, and routing layers, implementing VLAN segmentation, IP subnetting, trunking, router-on-a-stick methodology for inter-VLAN routing, DHCP services, wireless access points, and multiple network services including DNS, web, and email.

Technical Specifications

VLAN Architecture

v2 Enhancement: Added wireless devices (Laptop-CS, Laptop-ENG, AP-CS, AP-ENG) to VLANs and dedicated server VLAN with mail server.

VLAN ID	VLAN Name	Department	IP Subnet	Devices
10	CS_Dept	Computer Science	192.168.10.0/24	CS1, CS2, Laptop-CS, AP-CS
20	ENG_Dept	Engineering	192.168.20.0/24	ENG1, ENG2, Laptop-ENG, AP-ENG
30	BUS_Dept	Business	192.168.30.0/24	BUS1, BUS2
40	BIO_Dept	Biotechnology	192.168.40.0/24	BIO1, BIO2
99	SERVERS	Server Network	192.168.99.0/24	DNS1, MAIL1

IP Addressing Scheme

Device	VLAN	IP Address	Gateway	DNS Server
CS1	10	192.168.10.10	192.168.10.1	192.168.99.10
CS2	10	DHCP: 192.168.10.50+	192.168.10.1	192.168.99.10
ENG1	20	192.168.20.10	192.168.20.1	192.168.99.10
ENG2	20	DHCP: 192.168.20.50+	192.168.20.1	192.168.99.10
BUS1	30	192.168.30.10	192.168.30.1	192.168.99.10
BUS2	30	DHCP: 192.168.30.50+	192.168.30.1	192.168.99.10
BIO1	40	192.168.40.10	192.168.40.1	192.168.99.10
BIO2	40	DHCP: 192.168.40.50+	192.168.40.1	192.168.99.10
Laptop-CS	10	DHCP: 192.168.10.50+	192.168.10.1	192.168.99.10
Laptop-ENG	20	DHCP: 192.168.20.50+	192.168.20.1	192.168.99.10
DNS1	99	192.168.99.10	192.168.99.1	192.168.99.10
MAIL1	99	192.168.99.20	192.168.99.1	192.168.99.10

v2 Enhancement: Implemented DHCP addressing for client devices with mixed static/DHCP allocation strategy. Added MAIL1 server to server VLAN.

DHCP Configuration (v2 New Feature)

VLAN	DHCP Pool	IP Range	Gateway	Max Users
10 (CS_Dept)	VLAN10_CS	192.168.10.50-74	192.168.10.1	25
20 (ENG_Dept)	VLAN20_ENG	192.168.20.50-74	192.168.20.1	25
30 (BUS_Dept)	VLAN30_BUS	192.168.30.50-74	192.168.30.1	25
40 (BIO_Dept)	VLAN40_BIO	192.168.40.50-74	192.168.40.1	25

Wireless Network Configuration (v2 New Feature)

Mail Server Configuration (v2 New Feature)

- **Server:** MAIL1 (192.168.99.20)
- **Domain:** university.lan

Access Point	**SSID**	**Security**	**Password**
AP-CS (VLAN 10)	CS_Dept_WiFi	WPA2-PSK	CSdept2026
AP-ENG (VLAN 20)	ENG_Dept_WiFi	WPA2-PSK	ENGdept2026

- **Protocols:** SMTP (port 25), POP3 (port 110)
- **User Accounts:**
  - student1@university.lan / student123
  - professor1@university.lan / prof123
  - admin1@university.lan / admin123
- **Email Server Address:** mail.university.lan

Device Inventory and Physical Connections

- **Router:** 1 × Cisco 2911 (R1) *(v1)*
- **Switches:** 2 × Cisco 2960-24TT (SW1, SW2) *(v1)*
- **End Devices:** 8 × PCs *(v1)*, 2 × Laptops *(v2)*, 1 × DNS Server *(v1)*, 1 × Mail Server *(v2)*
- **Wireless Access:** 2 × AccessPoint-PT (AP-CS, AP-ENG) *(v2)*
- **Cabling:** Copper Straight-Through for all wired connections
- **Key Connections:**
  - R1 Gi0/0 SW1 Gi0/1 (Router to Switch - Trunk) *(v1)*
  - SW1 Gi0/2 SW2 Gi0/1 (Switch to Switch - Trunk) *(v1)*
  - DNS1 FastEthernet0 SW1 FastEthernet0/24 (Server - Access Port) *(v1)*
  - MAIL1 FastEthernet0 SW1 FastEthernet0/10 (Server - Access Port) *(v2)*
  - AP-CS FastEthernet0 SW1 FastEthernet0/8 (VLAN 10 Access) *(v2)*
  - AP-ENG FastEthernet0 SW1 FastEthernet0/9 (VLAN 20 Access) *(v2)*
  - Department PCs connected to designated switch ports as per VLAN assignments *(v1)*

Implementation Phases

Phase 1: Physical Topology Construction

Devices were placed in Cisco Packet Tracer workspace and connected according to the planned topology. Router R1 was connected to switch SW1 via GigabitEthernet ports, SW1 was connected to SW2 via trunk link, and all end devices were connected to their respective switches using FastEthernet access ports. **In v2**, wireless access points were added to provide WiFi connectivity for CS and Engineering departments.

Phase 2: Switch VLAN Configuration

VLANs were created on both switches with appropriate names and IDs (10: CS_Dept, 20: ENG_Dept, 30: BUS_Dept, 40: BIO_Dept, 99: SERVERS). Switch ports were configured as access ports and assigned to their corresponding VLANs based on connected devices. Trunk ports were configured between switches (SW1 Gi0/2 SW2 Gi0/1) and between SW1 and the router (SW1 Gi0/1 R1 Gi0/0). **In v2**, access point ports were configured in their respective departmental VLANs.

Phase 3: Router Inter-VLAN Routing

Router R1 was configured using router-on-a-stick methodology with subinterfaces created for each VLAN. Each subinterface was configured with 802.1Q encapsulation and assigned an IP address serving as the default gateway for devices in that VLAN (192.168.10.1 for VLAN 10, 192.168.20.1 for VLAN 20, 192.168.30.1 for VLAN 30, 192.168.40.1 for VLAN 40, and 192.168.99.1 for VLAN 99). **In v2**, DHCP relay agent was configured on all subinterfaces to forward DHCP requests to the central DHCP server.

Phase 4: Network Services Configuration (v2 Enhanced)

In v1, basic static IP addressing and DNS were configured. **In v2**, DHCP service was configured on DNS1 server with separate pools for each VLAN, providing automatic IP address assignment to client devices. DNS service was enhanced with additional records for new devices and services. Mail server (MAIL1) was configured with user accounts and integrated with DNS. Wireless access points were configured with WPA2 security and departmental SSIDs. Web services were configured on DNS1 server to host project documentation accessible via `http://university.lan`.

Testing and Results

Testing Methodology

Comprehensive connectivity tests were performed including intra-VLAN communication (same department), inter-VLAN routing (cross-department), gateway accessibility, **in v2:** DHCP address assignment, DNS resolution, wireless connectivity, email services, and server access from all departments. Both ping commands, web access, and email clients were used for validation across wired and wireless connections.

Verification Methods

Network verification was performed using standard Cisco IOS commands to check VLAN configurations (`show vlan brief`), trunk status (`show interfaces trunk`), router interface status (`show ip interface brief`), **in v2:** DHCP relay configuration, and connectivity between devices. DHCP lease tables, DNS resolution tests, and email send/receive tests were conducted to validate all network services.

Results

All tests were successfully completed, demonstrating proper network functionality including successful intra-VLAN and inter-VLAN communication, **in v2:** correct DHCP address assignment across all VLANs, functioning DNS name resolution for all devices and services, secure wireless connectivity with WPA2 encryption, working email services with cross-department communication, and effective network segmentation. All network services were accessible from both wired and wireless clients across different VLANs, confirming end-to-end network functionality.

Conclusion

This project successfully implemented a comprehensive multi-department LAN using Cisco Packet Tracer, demonstrating practical application of VLAN technology, inter-VLAN routing, **and in v2:** DHCP services, wireless networking, and multiple network services. The implementation followed concepts from existing research on LAN design and simulation, providing a functional model for departmental network segmentation in a modern educational institution. The project validated key networking concepts including VLAN segmentation, IP subnetting, trunking, router-on-a-stick configuration, **and in v2:** DHCP implementation with relay agents, wireless network integration, DNS and email service deployment, and comprehensive network testing. This implementation provides a robust foundation for more complex network designs and serves as an educational model for enterprise network architecture principles.